Muhammad Hamza Nawaz

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Profile

First-year M.S. student at Kyung Hee University, working on **intelligent metasurface-assisted ISAC**. I specialize in **deep learning and convex optimization** and have extensive experience with **applications of deep learning and optimization theory to wireless communication systems**, including MIMO-OFDM channel prediction, sum-rate and max-min-rate optimization for ISAC systems. I have collaborated on funded research with groups at the University of Quebec, and Mid Sweden University, Sweden. My research has resulted in **several peer-reviewed publications** in leading IEEE venues.

Education

- **M.S. in Electronics & Information Convergence Engineering** — Yongin, Korea
  Area of research: Stacked intelligent metasurface-assisted monostatic ISAC — *Mar. 2026 – Present*
  CGPA: 4.3/4.3, Supervisor: Yun Hee Kim
- **Bachelor of Engg. in Electrical Engg. | National University of Sciences and Technology** — Islamabad, Pakistan
  Area of research: Low-Complexity Deep-Learning architectures for Breathing Signals Classification — *Oct. 2020 – Jul. 2024*
  CGPA: 3.32/4, Project advisor: Ahmad Salman

Experience

- **Intelligent Communications Systems Lab, Kyung Hee University**
  Graduate Research Assistant — *Mar. 2026 - Present*
  Project: Intelligent Metasurface-Assisted ISAC
- **Laboratory of Signal and System Integration, Université du Québec à Trois-Rivières**
  Research Assistant — *May 2025 - Dec 2025*
  Project: Continual With Uncertainty Weighting for MIMO-OFDM Channel Prediction
- **Information Processing and Transmission Lab, NUST**
  Research Assistant — *May 2024 - Dec. 2024*
  Project: Safety-Critical 5G URLLC for Industrial IoT
- **Optimization and Machine Learning (OPTIMAL) Lab, NUST**
  Research Assistant — *May 2023 - May 2024*
  Project: Deep Learning Architectures for Signal Denoising and Classification

Projects

- **Intelligent Metasurface-Assisted ISAC** | MATLAB, CVX, Successive Convex Approximation (SCA), Gradient-Based Optimization, Non-Linear Optimization, Wave-Domain Beamforming
  - Proposed an RSMA-based SIM-assisted downlink ISAC, optimized using **block-coordinate gradient ascent**, which surpassed SDMA baseline to achieve **33% higher sum rate** while meeting beampattern gain constraint.
  - Extended the above work to **max-min fairness design** to achieve **8% improvement in min-rate** as compared to SDMA baseline.
  - Collaborated on RSMA-enabled STAR-RIS-assisted downlink ISAC to propose an SCA-based algorithm that achieved **19% higher sum-rate** as compared to SDMA baseline while meeting beampattern gain constraint.
  - Publications: [1, 2, 3]
- **Uncertainty-Aware Continual Learning for MIMO Channel Prediction** | Python, PyTorch, MATLAB (5G Toolbox), 3GPP TR 38.901, LSTM, Monte-Carlo Dropout, Continual Learning
  - Generated a **3GPP TR 38.901 CDL-D** 8×2 MIMO-OFDM CSI dataset in MATLAB with Doppler-induced channel aging to emulate non-stationary channels.
  - Built a **PyTorch LSTM predictor** with **Monte-Carlo dropout** for uncertainty estimation, trained with a **heteroscedastic uncertainty-weighted loss**.
  - Proposed **uncertainty-weighted experience replay** with LARS buffer replacement, achieving **flat per-RB NMSE** using replay memory **under 5% of the dataset**.
  - Publication: [4]

- **Safety-Critical 5G URLLC for Industrial IoT** | MATLAB (5G Toolbox), 3GPP NR PHY/MAC, HARQ, Configured Grant, Stochastic Modeling
  - Modeled a 5G NR industrial uplink with **configured-grant** routine traffic and **slot-level preemptive access** for safety-critical devices, in collaboration with Mid Sweden University and Manchester Metropolitan University.
  - Derived a **closed-form end-to-end safety response time model** (IEC 61784-3-3) linking HARQ retransmissions, BLER, and slot alignment to system-level delay.
  - Implemented custom **HARQ entities** in MATLAB 5G Toolbox, showing **over 50% lower response time** as subcarrier spacing increases from 15 to 60 KHz.
- **Deep Learning Architectures for Signal Denoising and Classification** | Python, CNN, Autoencoder, GAN, Attention, short-time Fourier transform (STFT), Transfer Learning
  - Designed a lightweight **1D CNN** on Mel-spectrograms for 7-class lung disease classification, achieving **99.9% accuracy** on ICBHI 2017 and surpassing the best reported results.
  - Built a **U-Net-inspired denoising autoencoder** and STFT-based spectral masking, then applied **transfer learning** to a self-collected dataset, achieving **95.5% accuracy** in a real-time mobile pipeline.
  - Proposed a **GAN** with **spatial-channel attention** generators for denoising, achieving **2 dB higher PSNR** with **2M fewer parameters**.
  - Publication: [5]
- **ResNet-Enhanced Image Classifier** | Python, TensorFlow, ResNet, CNN, Data Preprocessing
  - Developed an image classifier for diverse image datasets using a **ResNet-inspired architecture** in TensorFlow.
  - Implemented novel **data preprocessing** techniques to improve training from scratch.
  - Surpassed state-of-the-art from-scratch training accuracy on **CIFAR-100** by **5%**.
- **ColorGAN: Image Colorization** | Python, GAN, U-Net, CNN
  - Developed a novel **GAN-based architecture** for image colorization.
  - Designed a **U-Net generator** and a **CNN discriminator**.
  - Achieved results comparable to the benchmark with **10× fewer training epochs**.
- **Enhanced Audio: Spectral Denoising** | Python, Librosa, STFT, Statistical Signal Processing
  - Developed an audio denoising algorithm in Python (Librosa) using **frequency-domain analysis**.
  - Estimated per-frequency **noise thresholds** through statistical analysis of the noise spectrum.
  - Applied **STFT-based spectral masking** to suppress noise in noisy signals.

Publications

[1] J. Ahmad, **M. H. Nawaz**, and Y. H. Kim, "RSMA-Enabled Stacked Intelligent Metasurface-Assisted Downlink ISAC," (Accepted, Proc. IEEE Global Commun. Conf. (GLOBECOM), Macau, China, Dec. 2026.)

[2] **M. H. Nawaz**, J. Ahmad, M.A. Khan, and Y. H. Kim, "Max–Min Fairness Design for Stacked Intelligent Metasurface-Assisted RSMA-ISAC Systems," (Accepted: IEEE International Conference on ICT Convergence (ICTC), Jeju Island, Korea, Oct., 2026.)

[3] N. Khan, **M. H. Nawaz**, J. Ahmad, and Y. H. Kim, "Active STAR-RIS Empowered ISAC with RSMA Under Nonlinear Amplification," (Under Review: IEEE Trans. Veh. Technol.)

[4] M. J. Qamar, **M. H. Nawaz**, M. A. Ouameur, A. Mohsin, and M. Bagaa, "Uncertainty-Weighted Experience Replay for Continual MIMO Channel Prediction," Proc. IEEE International Conference on Communications (ICC), Glasgow, UK, May 2026. [PDF].

[5] **M. H. Nawaz**, J. Ahmad, M. Haroon, M. Haseeb, and A. Salman, "Real-Time Deep Learning for Lung Disease Classification: A Step Forward," Proc. International Conference on Frontiers of Information Technology (FIT), Islamabad, Dec. 2024. [PDF].

[6] **M. H. Nawaz**, J. Ahmad, M. A. Mohsin, and M. A. Jamshed, "Generative AI for Ambient Intelligence," (Accepted: IET Book Chapter.)

[7] M. A. Mohsin, J. Ahmad, **M. H. Nawaz**, and M. A. Jamshed, "Towards 6G Intelligence: The Role of Generative AI in Future Wireless Networks," (Under Review: Elsevier Book Chapter.) [PDF].

Teaching

- **Teaching Assistant**, School of Electrical Engineering and Computer Science, NUST *Sep. 2024 – Jan. 2025*
  - **Mobile Communication Systems (Fall 2024)**: conducted tutorials, graded assignments, quizzes, and exams, and assisted in preparing course materials.

Relevant Coursework

- **Communications Engineering:** Mathematics for Communications-I, Optimization Theory, Signals and Systems, Digital Signal Processing, Mobile Communication Systems, Electromagnetic Field Theory, Communication Systems, Computer & Communication Networks
- **Computer Science:** Programming Fundamentals, Object Oriented Programming, Data Structures and Algorithms, Computer Vision, Machine Learning

Awards

- Received the **Brain Korea 21 (BK21) FOUR Research Award**, the South Korean government's premier graduate research scholarship, awarded as a monthly stipend for the master's degree.
- Received the **President's Scholarship**, a highly competitive award covering full tuition for master's students at Kyung Hee University.
- Received the **NUST Community Impact Honor** for outstanding community service during undergraduate studies, NUST, Pakistan.
- Awarded the **Pakistan Engineering Council's Final Year Project Fund** for novelty and excellent performance in the final year design project.

Professional Activities

- TPC Member and Reviewer for IEEE Global Communications Conference (GLOBECOM) Workshops (2025, 2026)

Skills

AI	PyTorch, TensorFlow, OpenCV, CNNs, LSTMs, GANs, Autoencoders, Attention Nets
Programming	Python, C/C++, MATLAB, LaTeX
Wireless	Sionna, MATLAB 5G Toolbox, NYUSIM, QuaDRiGa
Optimization	CVX, SCA, Gradient-Based and Non-Linear Optimization
Data	NumPy, SciPy, Pandas, Matplotlib
Tools	Overleaf, Visual Studio, Google Colab, Proteus Suite

Leadership & Activities

- **Team Captain:** Captained the SEECS cricket team to victory in the inter-department tournament (2023).
- **Community Service:** Organized fundraising and a visit to an orphanage to support orphaned children.
- **Social Media Team Lead:** Managed social media coverage of all NUST Robotics Club events (2022–23).